1. Write a Pandas program to get the powers of an array values element-wise.

Note: First array elements raised to powers from second array

Sample data: {'X':[78,85,96,80,86], 'Y':[84,94,89,83,86],'Z':[86,97,96,72,83]}

Expected Output:

X Y Z

0 78 84 86

1 85 94 97

2 96 89 96

3 80 83 72

4 86 86 83

2. Write a Pandas program to create and display a DataFrame from a specified dictionary data which has the index labels.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

.... i 2 Kevin no 8.0

j 1 Jonas yes 19.0

3. Write a Pandas program to display a summary of the basic information about a specified DataFrame and its data.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Summary of the basic information about this DataFrame and its data:

<class 'pandas.core.frame.DataFrame'>

Index: 10 entries, a to j

Data columns (total 4 columns):

.... dtypes: float64(1), int64(1), object(2)

memory usage: 400.0+ bytes

None

4. Write a Pandas program to get the first 3 rows of a given DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

First three rows of the data frame:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

c 2 Katherine yes 16.5

5. Write a Pandas program to select the 'name' and 'score' columns from the following DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Select specific columns:

name score

a Anastasia 12.5

b Dima 9.0

c Katherine 16.5

... h Laura NaN

i Kevin 8.0

j Jonas 19.0

6. Write a Pandas program to select the specified columns and rows from a given data frame.

Sample Python dictionary data and list labels:

Select 'name' and 'score' columns in rows 1, 3, 5, 6 from the following data frame.

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Select specific columns and rows:

score qualify

b 9.0 no

d NaN no

f 20.0 yes

g 14.5 yes

7. Write a Pandas program to select the rows where the number of attempts in the examination is greater than 2.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Number of attempts in the examination is greater than 2:

name score attempts qualify

b Dima 9.0 3 no

d James NaN 3 no

f Michael 20.0 3 yes

8. Write a Pandas program to count the number of rows and columns of a DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Number of Rows: 10

Number of Columns: 4

9. Write a Pandas program to select the rows where the score is missing, i.e. is NaN.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Rows where score is missing:

attempts name qualify score

d 3 James no NaN

h 1 Laura no NaN

10. Write a Pandas program to select the rows the score is between 15 and 20 (inclusive).

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Rows where score between 15 and 20 (inclusive):

attempts name qualify score

c 2 Katherine yes 16.5

f 3 Michael yes 20.0

j 1 Jonas yes 19.0

11. Write a Pandas program to select the rows where number of attempts in the examination is less than 2 and score greater than 15.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Number of attempts in the examination is less than 2 and score greater than 15 :

name score attempts qualify

j Jonas 19.0 1 yes

12. Write a Pandas program to change the score in row 'd' to 11.5.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Change the score in row 'd' to 11.5:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

c 2 Katherine yes 16.5

...

i 2 Kevin no 8.0

j 1 Jonas yes 19.0

13. Write a Pandas program to calculate the sum of the examination attempts by the students.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Sum of the examination attempts by the students:

19

14. Write a Pandas program to calculate the mean score for each different student in DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Mean score for each different student in data frame:

13.5625

15. Write a Pandas program to append a new row 'k' to data frame with given values for each column. Now delete the new row and return the original DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Values for each column will be:

name : "Suresh", score: 15.5, attempts: 1, qualify: "yes", label: "k"

Expected Output:

Append a new row:

Print all records after insert a new record:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

......

j 1 Jonas yes 19.0

k 1 Suresh yes 15.5

Delete the new row and display the original rows:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

....

i 2 Kevin no 8.0

j 1 Jonas yes 19.0

16. Write a Pandas program to sort the DataFrame first by 'name' in descending order, then by 'score' in ascending order.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Values for each column will be:

name : "Suresh", score: 15.5, attempts: 1, qualify: "yes", label: "k"

Expected Output:

Orginal rows:

name score attempts qualify

a Anastasia 12.5 1 yes

b Dima 9.0 3 no

c Katherine 16.5 2 yes

d James NaN 3 no

e Emily 9.0 2 no

f Michael 20.0 3 yes

g Matthew 14.5 1 yes

h Laura NaN 1 no

i Kevin 8.0 2 no

j Jonas 19.0 1 yes

Sort the data frame first by 'name' in descending order, then by 'score' in ascending order:

name score attempts qualify

a Anastasia 12.5 1 yes

b Dima 9.0 3 no

c Katherine 16.5 2 yes

d James NaN 3 no

e Emily 9.0 2 no

f Michael 20.0 3 yes

g Matthew 14.5 1 yes

h Laura NaN 1 no

i Kevin 8.0 2 no

j Jonas 19.0 1 yes

17. Write a Pandas program to replace the 'qualify' column contains the values 'yes' and 'no' with True and False.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Replace the 'qualify' column contains the values 'yes' and 'no' with T

rue and False:

attempts name qualify score

a 1 Anastasia True 12.5

b 3 Dima False 9.0

......

i 2 Kevin False 8.0

j 1 Jonas True 19.0

18. Write a Pandas program to change the name 'James' to 'Suresh' in name column of the DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Change the name 'James' to \?Suresh\?:

attempts name qualify score

a 1 Anastasia yes 12.5

b 3 Dima no 9.0

....

i 2 Kevin no 8.0

j 1 Jonas yes 19.0

19. Write a Pandas program to delete the 'attempts' column from the DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

Delete the 'attempts' column from the data frame:

name qualify score

a Anastasia yes 12.5

b Dima no 9.0

.....

i Kevin no 8.0

j Jonas yes 19.0

20. Write a Pandas program to insert a new column in existing DataFrame.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

New DataFrame after inserting the 'color' column

attempts name qualify score color

a 1 Anastasia yes 12.5 Red

b 3 Dima no 9.0 Blue

....

i 2 Kevin no 8.0 Green

j 1 Jonas yes 19.0 Red

21. Write a Pandas program to iterate over rows in a DataFrame.

Sample Python dictionary data and list labels:

exam_data = [{'name':'Anastasia', 'score':12.5}, {'name':'Dima','score':9}, {'name':'Katherine','score':16.5}]

Expected Output:

Anastasia 12.5

Dima 9.0

Katherine 16.5

22. Write a Pandas program to get list from DataFrame column headers.

Sample Python dictionary data and list labels:

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],

'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Expected Output:

['attempts', 'name', 'qualify', 'score']

23. Write a Pandas program to rename columns of a given DataFrame

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 9

New DataFrame after renaming columns:

Column1 Column2 Column3

0 1 4 7

1 2 5 8

2 3 6 9

24. Write a Pandas program to select rows from a given DataFrame based on values in some columns.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

Rows for colum1 value == 4

col1 col2 col3

1 4 5 8

3 4 7 0

25. Write a Pandas program to change the order of a DataFrame columns.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

After altering col1 and col3

col3 col2 col1

0 7 4 1

1 8 5 4

2 9 6 3

3 0 7 4

4 1 8 5

26. Write a Pandas program to add one row in an existing DataFrame.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

After add one row:

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

5 10 11 12

27. Write a Pandas program to write a DataFrame to CSV file using tab separator.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

Data from new_file.csv file:

col1\tcol2\tcol3

0 1\t4\t7

1 4\t5\t8

2 3\t6\t9

3 4\t7\t0

4 5\t8\t1

28. Write a Pandas program to count city wise number of people from a given of data set (city, name of the person).

Sample data:

city Number of people

0 California 4

1 Georgia 2

2 Los Angeles 4

29. Write a Pandas program to delete DataFrame row(s) based on given column value.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

New DataFrame

col1 col2 col3

0 1 4 7

2 3 6 9

3 4 7 0

4 5 8 1

30. Write a Pandas program to widen output display to see more columns.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

31. Write a Pandas program to select a row of series/dataframe by given integer index.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

Index-2: Details

col1 col2 col3

2 3 6 9

32. Write a Pandas program to replace all the NaN values with Zero's in a column of a dataframe.

Sample data:

Original DataFrame

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

New DataFrame replacing all NaN with 0:

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no 0.0

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no 0.0

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

33. Write a Pandas program to convert index in a column of the given dataframe.

Sample data:

Original DataFrame

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

After converting index in a column:

index attempts name qualify score

0 0 1 Anastasia yes 12.5

1 1 3 Dima no 9.0

2 2 2 Katherine yes 16.5

3 3 3 James no NaN

4 4 2 Emily no 9.0

5 5 3 Michael yes 20.0

6 6 1 Matthew yes 14.5

7 7 1 Laura no NaN

8 8 2 Kevin no 8.0

9 9 1 Jonas yes 19.0

Hiding index:

index attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

34. Write a Pandas program to set a given value for particular cell in DataFrame using index value.

Sample data:

Original DataFrame

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

Set a given value for particular cell in the DataFrame

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 10.2

9 1 Jonas yes 19.0

35. Write a Pandas program to count the NaN values in one or more columns in DataFrame.

Sample data:

Original DataFrame

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

Number of NaN values in one or more columns:

2

36. Write a Pandas program to drop a list of rows from a specified DataFrame.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 4 5 8

2 3 6 9

3 4 7 0

4 5 8 1

New DataFrame after removing 2nd & 4th rows:

col1 col2 col3

0 1 4 7

1 4 5 8

3 4 7 0

37. Write a Pandas program to reset index in a given DataFrame.

Sample data:

Original DataFrame

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

After removing first and second rows

attempts name qualify score

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

Reset the Index:

index attempts name qualify score

0 2 2 Katherine yes 16.5

1 3 3 James no NaN

2 4 2 Emily no 9.0

3 5 3 Michael yes 20.0

4 6 1 Matthew yes 14.5

5 7 1 Laura no NaN

6 8 2 Kevin no 8.0

7 9 1 Jonas yes 19.0

38. Write a Pandas program to divide a DataFrame in a given ratio.

Sample data:

Original DataFrame:

0 1

0 0.316147 -0.767359

1 -0.813410 -2.522672

2 0.869615 1.194704

3 -0.892915 -0.055133

4 -0.341126 0.518266

5 1.857342 1.361229

6 -0.044353 -1.205002

7 -0.726346 -0.535147

8 -1.350726 0.563117

9 1.051666 -0.441533

70% of the said DataFrame:

0 1

8 -1.350726 0.563117

2 0.869615 1.194704

5 1.857342 1.361229

6 -0.044353 -1.205002

3 -0.892915 -0.055133

1 -0.813410 -2.522672

0 0.316147 -0.767359

30% of the said DataFrame:

0 1

4 -0.341126 0.518266

7 -0.726346 -0.535147

9 1.051666 -0.441533

39. Write a Pandas program to combining two series into a DataFrame.

Sample data:

Data Series:

0 100

1 200

2 python

3 300.12

4 400

dtype: object

0 10

1 20

2 php

3 30.12

4 40

dtype: object

New DataFrame combining two series:

0 1

0 100 10

1 200 20

2 python php

3 300.12 30.12

4 400 40

40. Write a Pandas program to shuffle a given DataFrame rows.

Sample data:

Original DataFrame:

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

2 2 Katherine yes 16.5

3 3 James no NaN

4 2 Emily no 9.0

5 3 Michael yes 20.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

New DataFrame:

attempts name qualify score

5 3 Michael yes 20.0

0 1 Anastasia yes 12.5

9 1 Jonas yes 19.0

6 1 Matthew yes 14.5

7 1 Laura no NaN

1 3 Dima no 9.0

3 3 James no NaN

4 2 Emily no 9.0

8 2 Kevin no 8.0

2 2 Katherine yes 16.5

41. Write a Pandas program to convert DataFrame column type from string to datetime.

Sample data:

String Date:

0 3/11/2000

1 3/12/2000

2 3/13/2000

dtype: object

Original DataFrame (string to datetime):

0

0 2000-03-11

1 2000-03-12

2 2000-03-13

42. Write a Pandas program to rename a specific column name in a given DataFrame.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 9

New DataFrame after renaming second column:

col1 Column2 col3

0 1 4 7

1 2 5 8

2 3 6 9

43. Write a Pandas program to get a list of a specified column of a DataFrame.

Sample data:

Powered by

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 9

Col2 of the DataFrame to list:

[4, 5, 6]

44. Write a Pandas program to create a DataFrame from a Numpy array and specify the index column and column headers.

Sample Output:

Column1 Column2 Column3

Index1 0 0.0 0.0

Index2 0 0.0 0.0

Index3 0 0.0 0.0

....

Index12 0 0.0 0.0

Index13 0 0.0 0.0

Index14 0 0.0 0.0

Index15 0 0.0 0.0

45. Write a Pandas program to find the row for where the value of a given column is maximum.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

Row where col1 has maximum value:

4

Row where col2 has maximum value:

3

Row where col3 has maximum value:

2

46. Write a Pandas program to check whether a given column is present in a DataFrame or not.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

Col4 is not present in DataFrame.

Col1 is present in DataFrame.

47. Write a Pandas program to get the specified row value of a given DataFrame.

Sample data:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

Value of Row

col1 1

col2 4

col3 7

Name: 0, dtype: int64

Value of Row4

col1 4

col2 9

col3 1

Name: 3, dtype: int64

48. Write a Pandas program to get the datatypes of columns of a DataFrame.

Sample data:

Original DataFrame:

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

....

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

Data types of the columns of the said DataFrame:

attempts int64

name object

qualify object

score float64

dtype: object

49. Write a Pandas program to append data to an empty DataFrame.

Sample data:

Original DataFrame:

After appending some data:

col1 col2

0 0 0

1 1 1

2 2 2

50. Write a Pandas program to sort a given DataFrame by two or more columns.

Sample data:

Original DataFrame:

attempts name qualify score

0 1 Anastasia yes 12.5

1 3 Dima no 9.0

....

8 2 Kevin no 8.0

9 1 Jonas yes 19.0

Sort the above DataFrame on attempts, name:

attempts name qualify score

0 1 Anastasia yes 12.5

9 1 Jonas yes 19.0

7 1 Laura no NaN

6 1 Matthew yes 14.5

4 2 Emily no 9.0

2 2 Katherine yes 16.5

8 2 Kevin no 8.0

1 3 Dima no 9.0

3 3 James no NaN

5 3 Michael yes 20.0

51. Write a Pandas program to convert the datatype of a given column (floats to ints).

Sample data:

Original DataFrame:

attempts name qualify score

0 1 Anastasia yes 12.50

1 3 Dima no 9.10

......

8 2 Kevin no 8.80

9 1 Jonas yes 19.13

Data types of the columns of the said DataFrame:

attempts int64

name object

qualify object

score float64

dtype: object

Now change the Data type of 'score' column from float to int:

attempts name qualify score

0 1 Anastasia yes 12

1 3 Dima no 9

2 2 Katherine yes 16

3 3 James no 12

4 2 Emily no 9

5 3 Michael yes 20

6 1 Matthew yes 14

7 1 Laura no 11

8 2 Kevin no 8

9 1 Jonas yes 19

Data types of the columns of the DataFrame now:

attempts int64

name object

qualify object

score int64

dtype: object

52. Write a Pandas program to remove infinite values from a given DataFrame.

Sample data:

Original DataFrame:

0

0 1000.000000

1 2000.000000

2 3000.000000

3 -4000.000000

4 inf

5 -inf

Removing infinite values:

0

0 1000.0

1 2000.0

2 3000.0

3 -4000.0

4 NaN

5 NaN

53. Write a Pandas program to insert a given column at a specific column index in a DataFrame.

Sample data:

Original DataFrame

col2 col3

0 4 7

1 5 8

2 6 12

3 9 1

4 5 11

New DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

54. Write a Pandas program to convert a given list of lists into a Dataframe.

Sample data:

Original list of lists:

[[2, 4], [1, 3]]

New DataFrame

col1 col2

0 2 4

1 1 3

55. Write a Pandas program to group by the first column and get second column as lists in rows.

Sample data:

Original DataFrame

col1 col2

0 C1 1

1 C1 2

2 C2 3

3 C2 3

4 C2 4

5 C3 6

6 C2 5

Group on the col1:

col1

C1 [1, 2]

C2 [3, 3, 4, 5]

C3 [6]

Name: col2, dtype: object

56. Write a Pandas program to get column index from column name of a given DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

Index of 'col2'

1

57. Write a Pandas program to count number of columns of a DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

Number of columns:

3

58. Write a Pandas program to select all columns, except one given column in a DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 8

2 3 6 12

3 4 9 1

4 7 5 11

All columns except 'col3':

col1 col2

0 1 4

1 2 5

2 3 6

3 4 9

4 7 5

59. Write a Pandas program to get first n records of a DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

First 3 rows of the said DataFrame':

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

60. Write a Pandas program to get last n records of a DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

Last 3 rows of the said DataFrame':

col1 col2 col3

3 4 9 12

4 7 5 1

5 11 0 11

61. Write a Pandas program to get topmost n records within each group of a DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

topmost n records within each group of a DataFrame:

col1 col2 col3

5 11 0 11

4 7 5 1

3 4 9 12

col1 col2 col3

3 4 9 12

2 3 6 8

1 2 5 5

4 7 5 1

col1 col2 col3

3 4 9 12

5 11 0 11

2 3 6 8

62. Write a Pandas program to remove first n rows of a given DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

After removing first 3 rows of the said DataFrame:

col1 col2 col3

3 4 9 12

4 7 5 1

5 11 0 11

63. Write a Pandas program to remove last n rows of a given DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

After removing last 3 rows of the said DataFrame:

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

64. Write a Pandas program to add a prefix or suffix to all columns of a given DataFrame.

Sample Output:

Original DataFrame

W X Y Z

0 68 78 84 86

1 75 85 94 97

2 86 96 89 96

3 80 80 83 72

4 66 86 86 83

Add prefix:

A_W A_X A_Y A_Z

0 68 78 84 86

1 75 85 94 97

2 86 96 89 96

3 80 80 83 72

4 66 86 86 83

Add suffix:

W_1 X_1 Y_1 Z_1

0 68 78 84 86

1 75 85 94 97

2 86 96 89 96

3 80 80 83 72

4 66 86 86 83

65. Write a Pandas program to reverse order (rows, columns) of a given DataFrame.

Sample Output:

Original DataFrame

W X Y Z

0 68 78 84 86

1 75 85 94 97

2 86 96 89 96

3 80 80 83 72

4 66 86 86 83

Reverse column order:

Z Y X W

0 86 84 78 68

1 97 94 85 75

2 96 89 96 86

3 72 83 80 80

4 83 86 86 66

Reverse row order:

W X Y Z

4 66 86 86 83

3 80 80 83 72

2 86 96 89 96

1 75 85 94 97

0 68 78 84 86

Reverse row order and reset index:

W X Y Z

0 66 86 86 83

1 80 80 83 72

2 86 96 89 96

3 75 85 94 97

4 68 78 84 86

66. Write a Pandas program to select columns by data type of a given DataFrame.

Sample Output:

Original DataFrame

name date_of_birth age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

Select numerical columns

age

0 18.5

1 21.2

2 22.5

3 22.0

4 23.0

Select string columns

name date_of_birth

0 Alberto Franco 17/05/2002

1 Gino Mcneill 16/02/1999

2 Ryan Parkes 25/09/1998

3 Eesha Hinton 11/05/2002

4 Syed Wharton 15/09/1997

67. Write a Pandas program to split a given DataFrame into two random subsets.

Sample Output:

Original Dataframe and shape:

name date_of_birth age

0 Alberto Franco 17/05/2002 18

1 Gino Mcneill 16/02/1999 21

2 Ryan Parkes 25/09/1998 22

3 Eesha Hinton 11/05/2002 22

4 Syed Wharton 15/09/1997 23

(5, 3)

Subset-1 and shape:

name date_of_birth age

1 Gino Mcneill 16/02/1999 21

4 Syed Wharton 15/09/1997 23

2 Ryan Parkes 25/09/1998 22

(3, 3)

Subset-2 and shape:

name date_of_birth age

0 Alberto Franco 17/05/2002 18

3 Eesha Hinton 11/05/2002 22

(2, 3)

68. Write a Pandas program to rename all columns with the same pattern of a given DataFrame.

Sample Output:

Original DataFrame

Name Date_Of_Birth Age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

Remove trailing (at the end) whitesapce and convert to lowercase of the columns name

name date_of_birth age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

69. Write a Pandas program to merge datasets and check uniqueness.

Sample Output:

Original DataFrame:

Name Date_Of_Birth Age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

New DataFrames:

Name Date_Of_Birth Age

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

Name Date_Of_Birth Age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

"one_to_one": check if merge keys are unique in both left and right datasets:"

Name Date_Of_Birth Age

0 Eesha Hinton 11/05/2002 22.0

1 Syed Wharton 15/09/1997 23.0

"one_to_many" or "1:m": check if merge keys are unique in left dataset:

Name Date_Of_Birth Age

0 Eesha Hinton 11/05/2002 22.0

1 Syed Wharton 15/09/1997 23.0

"any_to_one" or "m:1": check if merge keys are unique in right dataset:

Name Date_Of_Birth Age

0 Eesha Hinton 11/05/2002 22.0

1 Syed Wharton 15/09/1997 23.0

70. Write a Pandas program to convert continuous values of a column in a given DataFrame to categorical.

Input:

{ 'Name': ['Alberto Franco','Gino Mcneill','Ryan Parkes', 'Eesha Hinton', 'Syed Wharton'], 'Age': [18, 22, 40, 50, 80, 5] }

Output:

Age group:

0 kids

1 adult

2 elderly

3 adult

4 elderly

5 kids

Name: age_groups, dtype: category

Categories (3, object): [kids < adult < elderly]

71. Write a Pandas program to display memory usage of a given DataFrame and every column of the DataFrame.

Sample Output:

Original DataFrame:

Name Date_Of_Birth Age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Syed Wharton 15/09/1997 23.0

Global usage of memory of the DataFrame:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 5 entries, 0 to 4

Data columns (total 3 columns):

Name 5 non-null object

Date_Of_Birth 5 non-null object

Age 5 non-null float64

dtypes: float64(1), object(2)

memory usage: 801.0 bytes

None

The usage of memory of every column of the said DataFrame:

Index 80

Name 346

Date_Of_Birth 335

Age 40

dtype: int64

72. Write a Pandas program to combine many given series to create a DataFrame.

Sample Output:

Original Series:

0 php

1 python

2 java

3 c#

4 c++

dtype: object

0 1

1 2

2 3

3 4

4 5

dtype: int64

Combine above series to a dataframe:

index 0

0 1 python

1 2 java

2 3 c#

3 4 c++

4 5 NaN

Using pandas concat:

0 1

0 php 1

1 python 2

2 java 3

3 c# 4

4 c++ 5

Using pandas DataFrame with a dictionary, gives a specific name to the columns:

col1 col2

0 php 1

1 python 2

2 java 3

3 c# 4

4 c++ 5

73. Write a Pandas program to create DataFrames that contains random values, contains missing values, contains datetime values and contains mixed values.

Sample Output:

DataFrame: Contains random values:

A B C D

Dog2w4Dv4l 0.591058 1.883454 -1.608613 -0.502516

kV7mfdFcF9 0.629642 -0.474377 0.567357 1.658445

....

DataFrame: Contains missing values:

A B C D

i6i6Xn9l9c -0.299335 0.410871 -0.431840 -0.302177

OGo5KNNYNJ -0.174594 -1.366146 0.435063 -2.779446

u0mG9q1L7C 1.019094 -0.061077 -1.138138 -0.218460

RNJGqpci4o -0.380815 0.189970 -2.148521 -1.163589 vXIcxItZ1D NaN -0.079448 0.604777 0.065290

....

DataFrame: Contains datetime values:

A B C D

2000-01-03 0.665402 0.860808 -0.180986 -0.970889

2000-01-04 -1.511533 0.487539 -0.710355 -0.807816

2000-01-05 -0.773294 0.197918 -1.214035 1.049529

2000-01-06 -1.074894 1.774147 -0.620025 0.740779

....

DataFrame: Contains mixed values:

A B C D

0 0.0 0.0 foo1 2009-01-01

1 1.0 1.0 foo2 2009-01-02

2 2.0 0.0 foo3 2009-01-05

3 3.0 1.0 foo4 2009-01-06

4 4.0 0.0 foo5 2009-01-07

74. Write a Pandas program to fill missing values in time series data.

From Wikipedia , in the mathematical field of numerical analysis, interpolation is a type of estimation, a method of constructing new data points within the range of a discrete set of known data points.

Sample Output:

Original DataFrame:

c1 c2

2000-01-03 120.0 7.0

2000-01-04 130.0 NaN

2000-01-05 140.0 10.0

2000-01-06 150.0 NaN

2000-01-07 NaN 5.5

2000-01-10 170.0 16.5

DataFrame after interpolate:

c1 c2

2000-01-03 120.0 7.00

2000-01-04 130.0 8.50

2000-01-05 140.0 10.00

2000-01-06 150.0 7.75

2000-01-07 160.0 5.50

2000-01-10 170.0 16.50

75. Write a Pandas program to use a local variable within a query.

Sample Output:

Original DataFrame

W X Y Z

0 68 78 84 86

1 75 85 94 97

2 86 96 89 96

3 80 80 83 72

4 66 86 86 83

Values which are less than maximum value of 'W' column

W X Y Z

0 68 78 84 86

1 75 85 94 97

3 80 80 83 72

4 66 86 86 83

76. Write a Pandas program to clean object column with mixed data of a given DataFrame using regular expression.

Sample Output:

Original dataframe:

agent purchase

0 a001 4500

1 a002 7500

2 a003 $3000.25

3 a003 $1250.35

4 a004 9000.00

Data Types:

0 <class 'float'>

1 <class 'float'>

2 <class 'str'>

3 <class 'str'>

4 <class 'str'>

Name: purchase, dtype: object

New Data Types:

0 <class 'float'>

1 <class 'float'>

2 <class 'float'>

3 <class 'float'>

4 <class 'float'>

Name: purchase, dtype: object

77. Write a Pandas program to get the numeric representation of an array by identifying distinct values of a given column of a dataframe.

Sample Output:

Original DataFrame:

Name Date_Of_Birth Age

0 Alberto Franco 17/05/2002 18.5

1 Gino Mcneill 16/02/1999 21.2

2 Ryan Parkes 25/09/1998 22.5

3 Eesha Hinton 11/05/2002 22.0

4 Gino Mcneill 15/09/1997 23.0

Numeric representation of an array by identifying distinct values:

[0 1 2 3 1]

Index(['Alberto Franco', 'Gino Mcneill', 'Ryan Parkes', 'Eesha Hinton'], dtype='object')

78. Write a Pandas program to replace the current value in a dataframe column based on last largest value. If the current value is less than last largest value replaces the value with 0.

Test data:

rnum

0 23

1 21

2 27

3 22

...

10 34

11 19

12 31

13 32

14 19

Sample Output:

Original DataFrame:

rnum

0 23

1 21

2 27

3 22

...

10 34

11 19

12 31

13 32

14 19

Replace current value in a dataframe column based on last largest value:

rnum

0 23

1 0

2 27

3 0

...

10 34

11 0

12 0

13 0

14 0

79. Write a Pandas program to create a DataFrame from the clipboard (data from an Excel spreadsheet or a Google Sheet).

Sample Excel Data:

	A	B	C	D
1	1	2	3	4
2	2	3	4	5
3	4	5	1	0
4	2	3	7	8

Sample Output:

Data from clipboard to DataFrame:

1 2 3 4

0 2 3 4 5

1 4 5 1 0

2 2 3 7 8

80. Write a Pandas program to check for inequality of two given DataFrames.

Sample Output:

Original DataFrames:

W X Y Z

0 68.0 78.0 84 86

1 75.0 85.0 94 97

2 86.0 NaN 89 96

3 80.0 80.0 83 72

4 NaN 86.0 86 83

W X Y Z

0 78.0 78 84 86

1 75.0 85 84 97

2 86.0 96 89 96

3 80.0 80 83 72

4 NaN 76 86 83

Check for inequality of the said dataframes:

W X Y Z

0 True False False False

1 False False True False

2 False True False False

3 False False False False

4 True True False False

81. Write a Pandas program to get lowest n records within each group of a given DataFrame.

Sample Output:

Original DataFrame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

Lowest n records within each group of a DataFrame:

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

col1 col2 col3

5 11 0 11

0 1 4 7

1 2 5 5

col1 col2 col3

4 7 5 1

1 2 5 5

0 1 4 7